

Experiments: Part I

PS200C Quant Methods in Politics, Spring 2026

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Randomization Solves the Selection Problem

Recall this formula for the bias of the DIM ($\tilde{\tau}$) relative to ATT:

$$\begin{aligned}\tilde{\tau} &= \mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] \quad (\text{obs. diff. in means}) \\ &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \\ &= \underbrace{\mathbb{E}[Y_{1i} - Y_{0i}|D_i = 1]}_{\tau_{ATT}} + \underbrace{\mathbb{E}[Y_{0i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0]}_{\text{Bias}}\end{aligned}$$

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Stare at the bias term and ask: What could ensure it is zero?

Random assignment of D_i will make the treated and untreated units identical on average, such that

$$\mathbb{E}[Y_{0i} | D_i = 1] = \mathbb{E}[Y_{0i} | D_i = 0]$$

Historical Context

From “Experimentation and social interventions: a forgotten but important history” (Ann Oakley, 1998, *BMJ*). From the summary points:

- Many social scientists argue that randomised controlled trials are inappropriate for evaluating social interventions, but they ignore a considerable history, mainly in the United States, of the use of randomised controlled trials to assess different approaches to public policy and health promotion

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- This approach became less popular as policy makers reacted negatively to evidence of “near zero” effects
- Lessons to be learnt about implementing randomised controlled trials in real life settings include the difficulty of assessing complex multi-level interventions and the challenge of integrating qualitative data

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Re-rise of experiments

Though much of the history forgotten, renewed enthusiasm and “rediscovery” of experiments for social sciences early this century

- *Abbreviated* list of examples (from Green 2008):
 - **Program evaluation**: development programs, education programs, SAT prep classes, weight loss programs, diversity training, deliberative polls, advertising campaigns, website designs...
 - **Public policy evaluation**: teacher pay, student incentives, class size, speed traps, vouchers, alternative sentencing, job training, health insurance subsidies, tax compliance, public housing
 - **Behavioral research**: persuasion, mobilization, education, income, interpersonal influence, conscientious health behaviors, media exposure, deliberation, discrimination
 - **Research on institutions**: transparency, corruption, electoral systems, information
- Also 2019 Nobel prize in economics.

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Golden rule of inference: **Identification precedes estimation**

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For each we discuss:

- 1 the **identification assumption** first, which your life will depend upon when using this approach.
- 2 information from outside the data that help support that assumption
- 3 the (limited) things you can test in efforts to falsify the assumption
- 4 estimators
- 5 ideally, sensitivity to violations of assumptions

Classical Randomized Experiment

Setup:

- Units: $i = 1, \dots, N$
- Treatment: $D_i \in \{0, 1\}$, randomly assigned
- Potential outcomes: Y_{0i}, Y_{1i}
- Observed outcome: $Y_i = Y_{D_i i}$
- Number of treated/untreated units: $N_1 = \sum_{i=1}^N D_i$ and $N_0 = N - N_1$

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Notes:

- Random assignment can take one of several forms:
 - **Complete randomization**: Exactly N_1 treated units
 - **Simple (Bernoulli) randomization**: Each unit independently assigned to treatment with probability p

Identification of Average Treatment Effect Under Experiments

Identification Assumption: (guaranteed by random assignment):

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$$\begin{aligned} \mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}] &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \quad (\text{why?}) \\ &= \mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] \quad (\text{why?}) \\ &= \text{DIM Estimand} \\ &\approx \bar{Y}_1 - \bar{Y}_0 \quad (\text{the DIM estimator}) \end{aligned}$$

SATE and PATE

- Often we focus on estimating the average causal effect in a particular sample: **Sample Average Treatment Effect (SATE)**
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- Compare to the **Population Average Treatment Effect (PATE)**
 - Imagines uncertainty in both the sample drawn from the (larger) population, and in treatment assignment.
- If your sampling was random from a given population, then $\mathbb{E}[\text{SATE}] = \text{PATE}$. For this reason, we're often uninterested in distinguishing the two quantities.

Standard Errors for the ATE

Variance of the difference in means estimator:

$$\mathbb{V}(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma_{Y_1}^2}{N_1} + \frac{\sigma_{Y_0}^2}{N_0}$$

where $\sigma_{Y_1}^2$ and $\sigma_{Y_0}^2$ are the variance of the Y_{1i} and Y_{0i} in the population.

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Key facts:

- This is the correct SE for the **PATE**
- It is what you get from t-tests (unequal variance), or regression with robust SE
- It is **conservative** for the **SATE** (proof in appendix)

Example: Effect of Training on Earnings

- Treatment Group:
 - $N_1 = 7,487$
 - Estimated Average Earnings $\bar{Y}_1$: \$16,199
 - Estimated Sample Standard deviation $\hat{\sigma}_{Y|D_i=1}$: \$17,038
- Control Group :
 - $N_0 = 3,717$
 - Estimated Average Earnings $\bar{Y}_0$: \$15,040
 - Estimated Sample deviation $\hat{\sigma}_{Y|D_i=0}$: \$16,180
- Estimated average effect of training:

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 - $\hat{\tau}_{ATE} = \bar{Y}_1 - \bar{Y}_0 = 16,199 - 15,040 = \$1,159$
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$$\widehat{SE}_{\widehat{\tau}_{ATE}} = \sqrt{\frac{\hat{\sigma}_{Y|D_i=1}^2}{N_1} + \frac{\hat{\sigma}_{Y|D_i=0}^2}{N_0}} = \sqrt{\frac{17,038^2}{7,487} + \frac{16,180^2}{3,717}} \approx \$330$$

- Is this consistent with a zero average treatment effect $\alpha_{ATE} = 0$?

Testing the Null Hypothesis of Zero Average Effect

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- 1,159 is more than 2 SEs from 0, so you know t -stat/ z -score would be over 2 and p -value would be less than 0.05.

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- to recall the formalities,

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Use a two-sample t (or z) test with unequal variances:

$$t = \frac{\hat{\tau}}{\sqrt{\frac{\hat{\sigma}_{Y_i|D_i=1}^2}{N_1} + \frac{\hat{\sigma}_{Y_i|D_i=0}^2}{N_0}}} = \frac{\$1,159}{\sqrt{\frac{\$17,038^2}{7,487} + \frac{\$16,180^2}{3,717}}} \approx 3.5$$

- For large enough N , $t \xrightarrow{d} N(0, 1)$,
- Getting a 3.5 or above from a normal distribution would be quite a shock:

```
> 2 * (1 - pnorm(abs(3.5)))
[1] 0.000465
```

- Inverting the test statistic we can construct a 95% confidence interval

$$\hat{\tau}_{ATE} \pm 1.96 \cdot \widehat{SE}_{\widehat{ATE}}$$

- What assumptions did we need along the way?

Testing the Null Hypothesis of Zero Average Effect

```
> d <- read.dta("jtpa.dta")  
> t.test(earnings~assignmt,data=d,var.equal=FALSE)
```

Welch Two Sample t-test

```
data:  earnings by assignmt  
t = -3.5084, df = 7765.599, p-value = 0.0004533  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
  -1807.2427  -511.6239  
sample estimates:  
mean in group 0 mean in group 1  
   15040.50      16199.94
```

Regression to Estimate ATE in Experiments

Estimator (Regression with Experiments)

The ATE can be expressed as a regression coefficient:

$$\begin{aligned}
 Y_i &= D_i Y_{1i} + (1 - D_i) Y_{0i} \\
 &= Y_{0i} + (Y_{1i} - Y_{0i}) D_i \\
 &= \underbrace{\bar{Y}_0}_{\alpha} + \underbrace{(\bar{Y}_1 - \bar{Y}_0)}_{\tau_{Reg}} D_i + \underbrace{\{(Y_{i0} - \bar{Y}_0) + D_i \cdot [(Y_{i1} - \bar{Y}_1) - (Y_{i0} - \bar{Y}_0)]\}}_{\varepsilon} \\
 &= \alpha + \tau_{Reg} D_i + \varepsilon_i
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 \end{aligned}$$

- Does this assume constant treatment effects?
- Our SE estimator allows different variance for $D = 1$ and $D = 0$.
 - Implies heteroskedasticity
 - Use “HC2” heteroskedasticity-robust variance:

$$\hat{\sigma}_{HC2}^2 = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} = \tilde{V}(\tilde{\tau})$$

Regression to Estimate the Average Treatment Effect

```
> library(sandwich)
> library(lmtest)
>
> lout <- lm(earnings~assignmt,data=d)
> coeftest(lout,vcov = vcovHC(lout, type = "HC2"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	15040.50	265.38	56.6752	< 2.2e-16	***
assignmt	1159.43	330.46	3.5085	0.0004524	***
--					

Testing in Small Samples: Fisher's Exact Test

- Test of differences in means with large N :

$$H_0 : \mathbb{E}[Y_1] = \mathbb{E}[Y_0], \quad H_1 : \mathbb{E}[Y_1] \neq \mathbb{E}[Y_0] \text{ (weak null)}$$

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...which allows you to see many effect estimates all under this null.

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- 2 Obtain the null distribution of the statistic by calculating the same statistic $\hat{\theta}(\omega)$ under the sharp null for every possible (or many) ω in Ω
- 3 Compare $\hat{\theta}_{true}$ to the null distribution of $\hat{\theta}(\omega)$'s to see how "extreme" it is

Fisher's Exact Test: Setup

i	Y_{1i}	Y_{0i}	D_i
1	3	?	1
2	1	?	1
3	?	0	0
4	?	1	0
$\hat{\tau}_{ATE}$			1.5

What do we know given the sharp null $H_0 : Y_{1i} = Y_{0i}$?

Fisher's Exact Test: Sharp Null Fills In POs

i	Y_{1i}	Y_{0i}	D_i
1	3	3	1
2	1	1	1
3	0	0	0
4	1	1	0
$\hat{\tau}_{ATE}$			1.5
$\hat{\tau}(\omega)$			1.5

Given the full schedule of potential outcomes under the sharp null, we can compute the null distribution of the ATE across all possible randomizations.

Fisher's Exact Test: First Alternative Randomization

i	Y_{1i}	Y_{0i}	D_i	ω_1
1	3	3	1	1
2	1	1	1	0
3	0	0	0	1
4	1	1	0	0
$\hat{\tau}_{\text{ATE}}$			1.5	
$\hat{\tau}(\omega)$			1.5	0.5

Fisher's Exact Test: Adding Randomizations

i	Y_{1i}	Y_{0i}	D_i	ω_1	ω_2
1	3	3	1	1	1
2	1	1	1	0	0
3	0	0	0	1	0
4	1	1	0	0	1
$\hat{\tau}_{ATE}$				1.5	
$\hat{\tau}(\omega)$				1.5	0.5 1.5

Fisher's Exact Test: Adding Randomizations

i	Y_{1i}	Y_{0i}	D_i	ω_1	ω_2	ω_3
1	3	3	1	1	1	0
2	1	1	1	0	0	1
3	0	0	0	1	0	1
4	1	1	0	0	1	0
$\hat{\tau}_{ATE}$				1.5		
$\hat{\tau}(\omega)$				1.5	0.5	-1.5

Fisher's Exact Test: Adding Randomizations

i	Y_{1i}	Y_{0i}	D_i	ω_1	ω_2	ω_3	ω_4
1	3	3	1	1	1	0	0
2	1	1	1	0	0	1	1
3	0	0	0	1	0	1	0
4	1	1	0	0	1	0	1
$\hat{\tau}_{ATE}$				1.5			
$\hat{\tau}(\omega)$				1.5	0.5	1.5	-1.5
					-0.5		

Fisher's Exact Test: All Randomizations

i	Y_{1i}	Y_{0i}	D_i	ω_1	ω_2	ω_3	ω_4	ω_5
1	3	3	1	1	1	0	0	0
2	1	1	1	0	0	1	1	0
3	0	0	0	1	0	1	0	1
4	1	1	0	0	1	0	1	1
$\hat{\tau}_{ATE}$				1.5				
$\hat{\tau}(\omega)$				1.5	0.5	1.5	-1.5	-0.5

So *under null*: $\Pr(|\hat{\tau}(\omega)| \geq |\hat{\tau}_{ATE}|) = 4/6 \approx .67$.

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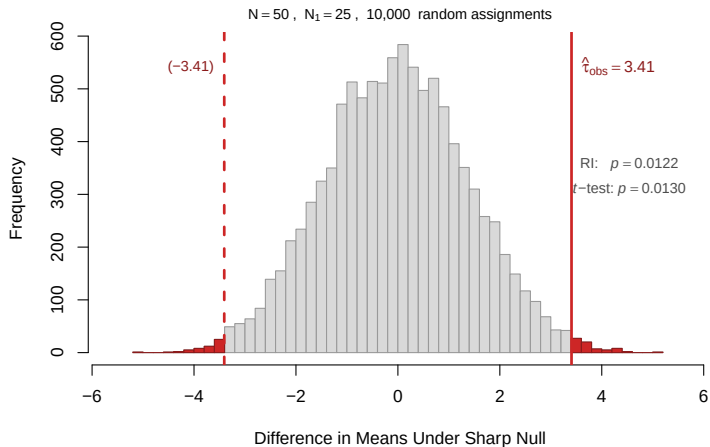
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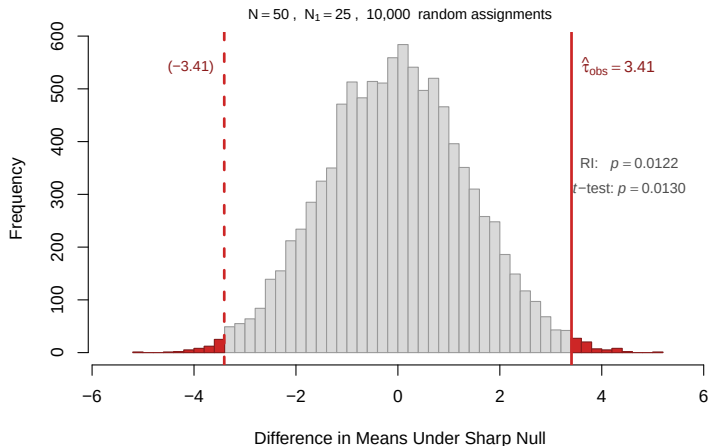
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- This gives us the **null distribution** of the test statistic
- Compare our observed $\hat{\tau}$ to this distribution

Randomization Inference in Practice



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Histogram: 10,000 sampled re-randomizations under sharp null.

Solid line: observed $\hat{t}$. Shaded regions: values at least as extreme $\Rightarrow$ p-value.

How do these inferences differ?

Finite-sample exactness.

- The Fisher/RI p-value is exact under the sharp null — no appeal to large- N approximations (CLT).
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Bottom line: RI valuable in **small samples** where normality is a worry.

Appendix

The exact SE for SATE

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$$SE_{\widehat{ATE}} = \sqrt{\left(\frac{N - N_1}{N - 1}\right) \frac{\sigma_{Y_1}^2}{N_1} + \left(\frac{N - N_0}{N - 1}\right) \frac{\sigma_{Y_0}^2}{N_0} + \left(\frac{1}{N - 1}\right) 2\text{Cov}[Y_1, Y_0]}$$

with population variances and covariances $\sigma_{Y_d}^2$, $\text{Cov}(Y_1, Y_0)$.

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We don't use this—can you see why?

Proof that Neyman SE is conservative for SATE

Two preliminaries:

- $\text{Cov}(Y_1, Y_0) \leq \text{SD}(Y_1)\text{SD}(Y_0)$
- $\text{Var}(Y_1) + \text{Var}(Y_0) \geq 2\sqrt{\text{Var}(Y_1)\text{Var}(Y_0)}$

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- $Var(Y_1) + Var(Y_0) \geq 2\sqrt{Var(Y_1)Var(Y_0)}$

With that,

$$\begin{aligned} SE_{\widehat{ATE}} &= \sqrt{\left(\frac{N - N_1}{N - 1}\right) \frac{Var[Y_1]}{N_1} + \left(\frac{N - N_0}{N - 1}\right) \frac{Var[Y_0]}{N_0} + \left(\frac{1}{N - 1}\right) 2Cov[Y_1, Y_0]} \\ &= \sqrt{\frac{1}{N - 1} \left(\frac{N_0}{N_1} Var[Y_1] + \frac{N_1}{N_0} Var[Y_0] + 2Cov[Y_1, Y_0] \right)} \\ &\leq \sqrt{\frac{1}{N - 1} \left(\frac{N_0}{N_1} Var[Y_1] + \frac{N_1}{N_0} Var[Y_0] + 2\sqrt{Var[Y_1]Var[Y_0]} \right)} \\ &\leq \sqrt{\frac{1}{N - 1} \left(\frac{N_0}{N_1} Var[Y_1] + \frac{N_1}{N_0} Var[Y_0] + Var[Y_1] + Var[Y_0] \right)} \end{aligned}$$

$SE_{\widehat{ATE}} \leq \widehat{SE}_{\widehat{ATE}}$, continued.

$$\begin{aligned}
 SE_{\widehat{ATE}} &\leq \sqrt{\frac{1}{N-1} \left(\frac{N_0}{N_1} \text{Var}[Y_1] + \frac{N_1}{N_0} \text{Var}[Y_0] + \text{Var}[Y_1] + \text{Var}[Y_0] \right)} \\
 &\leq \sqrt{\frac{N_0^2 \text{Var}[Y_1] + N_1^2 \text{Var}[Y_0] + N_1 N_0 (\text{Var}[Y_1] + \text{Var}[Y_0])}{(N-1)N_1 N_0}} \\
 &\leq \sqrt{\frac{(N_0^2 + N_1 N_0) \text{Var}[Y_1] + (N_1^2 + N_1 N_0) \text{Var}[Y_0]}{(N-1)N_1 N_0}} \\
 &\leq \sqrt{\frac{(N_0 + N_1)N_0 \text{Var}[Y_1]}{(N-1)N_1 N_0} + \frac{(N_1 + N_0)N_1 \text{Var}[Y_0]}{(N-1)N_1 N_0}} \\
 &\leq \sqrt{\frac{N \text{Var}[Y_1]}{(N-1)N_1} + \frac{N \text{Var}[Y_0]}{(N-1)N_0}} \\
 &\leq \sqrt{\frac{N}{N-1} \left(\frac{\text{Var}[Y_1]}{N_1} + \frac{\text{Var}[Y_0]}{N_0} \right)} \\
 &\leq \sqrt{\left(\frac{\text{Var}[Y_1]}{N_1} + \frac{\text{Var}[Y_0]}{N_0} \right)}
 \end{aligned}$$

$\therefore$ our estimator is conservative.